

**SIXTH SEMESTER (CUCBCSS—UG) DEGREE [SPECIAL] EXAMINATION
MARCH 2021**

Mathematics

MAT 6B 12—NUMBER THEORY AND LINEAR ALGEBRA

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes

Total No. of Questions : 20

Maximum : 20 Marks

INSTRUCTIONS TO THE CANDIDATE

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MAT 6B 12—NUMBER THEORY AND LINEAR ALGEBRA

(Multiple Choice Questions for SDE Candidates)

1. A subset W of a vector space V is a subspace if :
 - (A) The sum of elements of W belongs to W .
 - (B) The sum and scalar multiples of elements of W belongs to W .
 - (C) The scalar multiples of elements of W belongs to W .
 - (D) The sum of elements of V belongs to W .
2. The intersection of any set of subspaces of a vector space V is :
 - (A) Not a subspace of V .
 - (B) A subspace of V .
 - (C) Need not be a subspace of V .
 - (D) A proper subspace of V .
3. If B_1 and B_2 are any *two* finite bases of a vector space V then :
 - (A) Number of elements in $B_1 >$ number of elements in B_2 .
 - (B) Number of elements in $B_1 <$ number of elements B_2 .
 - (C) Number of elements in $B_1 =$ number of elements in B_2 .
 - (D) Number of elements in $B_1 \neq$ number of elements in B_2 .
4. If the map $f : V \rightarrow W$ is linear. Then the kernel or null space of f is :
 - (A) $f \rightarrow (W)$.
 - (B) $f \rightarrow (\{0_V\})$.
 - (C) $f \leftarrow (\{0_W\})$.
 - (D) $f \leftarrow (W)$.
5. Suppose a and b are integers with $a \neq 0$, then $a \mid b$ if :
 - (A) $a = bc, c$ is some integer.
 - (B) $b = ac, c$ is some integer.
 - (C) $c = ab, c$ is some integer.
 - (D) $ab = 1$.
6. $\gcd(-5, 5) = \underline{\hspace{2cm}}$.
 - (A) 3.
 - (B) 1.
 - (C) 5.
 - (D) -5.

7. Let a and b be integers, not both zero. Then a and b are relatively prime if and only if there exists integers x and y such that :
- (A) $1 = ax + by$. (B) $2 = ax + by$.
(C) $ab = ax + by$. (D) $a - b = ax + by$.
8. The number of primes is :
- (A) Finite. (B) Infinite.
(C) Uncountable. (D) 1729.
9. If $\gcd(a, b) = d$, then $\gcd(a/d, b/d) = \underline{\hspace{2cm}}$.
- (A) 1. (B) b .
(C) a . (D) d .
10. If a is an odd integer then $\gcd(3a, 3a + 2) = \underline{\hspace{2cm}}$.
- (A) 3. (B) 5.
(C) 1. (D) 2.
11. Every positive integer $n > 1$ can be expressed as a product of :
- (A) Composite numbers. (B) Prime numbers.
(C) Even numbers. (D) Odd numbers.
12. If p is a prime and $p|ab$, then :
- (A) $p|a$ only. (B) $p|b$ only.
(C) $p|a$ or $p|b$. (D) $p|a$ and $p|b$.
13. The Sieve of Eratosthenes is used for finding :
- (A) All primes below a given integer.
(B) All even numbers below a given integer.
(C) All odd numbers below a given integer.
(D) All composite numbers below a given integer.

Turn over

14. The system of linear congruences $ax + by \equiv r \pmod{n}$, $cx + dy \equiv s \pmod{n}$ has a unique solution modulo n whenever :
- (A) $\gcd(a - c, n) = 1$. (B) $\gcd(a - d, n) = 1$.
 (C) $\gcd(ad - bc, n) = 2$. (D) $\gcd(ad - bc, n) = 1$.
15. The solution of $25x \equiv 15 \pmod{29}$ is :
- (A) $x \equiv 18 \pmod{29}$. (B) $x \equiv 29 \pmod{29}$.
 (C) $x \equiv 18 \pmod{19}$. (D) $x \equiv 17 \pmod{19}$.
16. A number theoretic function f is said to be multiplicative if whenever $\gcd(m, n) = 1$, then :
- (A) $f(mn) = f(m) + f(n)$. (B) $f(mn) = f(m) f(n)$.
 (C) $f(mn) = f(m) - f(n)$. (D) $f(mn) = f(m)/f(n)$.
17. The domain of number theoretic function is the :
- (A) Set of prime numbers. (B) Set of negative integers.
 (C) Set of rational numbers. (D) Set of positive integers.
18. If n and r are positive integers with $1 \leq r < n$, then the binomial co-efficient $\binom{n}{r} = \frac{n!}{r!(n-r)!}$ is :
- (A) An integer. (B) A prime.
 (C) Irrational. (D) An even integer.
19. If n is a prime, then $\phi(n) = \underline{\hspace{2cm}}$
- (A) n . (B) $n - 1$.
 (C) $n + 1$. (D) $n - 2$.
20. For $n > 2$, $\phi(n)$ is :
- (A) An odd integer. (B) An even integer.
 (C) Irrational. (D) Prime.

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MARCH 2021**

Mathematics

MAT 6B 12—NUMBER THEORY AND LINEAR ALGEBRA

Time : Three Hours

Maximum 120 Marks

Section A

*Answer all questions.
Each question carries 1 mark.*

1. Define relatively prime integers. Give an example.
2. State the relationship between gcd and lcm of two positive integers.
3. Give an example for a Diophantine equation in two variables.
4. Write the canonical form of 720.
5. State Wilson's theorem.
6. Define an arithmetic multiplicative function.
7. Find $\phi(25)$.
8. Define subspace spanned by a non-empty subset of a vector space.
9. Give the canonical basis of $\mathbb{R}^n$.
10. Find the kernel of the differentiation map $D : \mathbb{R}_n[x] \rightarrow \mathbb{R}_n[x]$.
11. Define nullity of a linear map.
12. State the Dimension theorem.

(12 × 1 = 12 marks)

Section B

*Answer at least **eight** questions.
Each question carries 6 marks.
All questions can be attended.
Overall Ceiling 48.*

13. Show that the square of any odd positive integer is of the form $8k + 1$.
14. Prove that if $d|n$, then $(2^d - 1) | (2^n - 1)$.
15. Find gcd (143, 227).
16. Prove that there are infinite number of primes.

Turn over

17. Let a and b be arbitrary integers. Prove that $a \equiv b \pmod{n}$ if and only if a and b leave the same non-negative remainder when divided by n .
18. Find the units digit of 3^{100} .
19. Show that $28! + 233$ is divisible by 31.
20. Prove that $\phi(n) = n - 1$ if and only if n is prime.
21. Find the highest power of 5 dividing $1000!$.
22. Verify whether the set of all $n \times n$ symmetric matrices over a field F is a subspace of the space of all $n \times n$ matrices over F .
23. Prove that if S is a subset of a vector space V , then $\langle S \rangle = \text{span } S$.
24. Show that the set of vectors $\{(1, 0, -1), (1, 2, 1), (0, -3, 2)\}$ is a basis of $\mathbb{R}^3$.
25. Show that the mapping $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by $f(x, y) = (x + y, x - y)$ is linear.
26. Find the rank and nullity of the linear map $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ given by $f(x, y, z) = x$.

(8 × 6 = 48 marks)

Section C

Answer at least **five** questions.

Each question carries 9 marks.

All questions can be attended.

Overall Ceiling 45.

27. Let a and b be integers, not both zero. Prove that a and b are relatively prime if and only if there exist integers x and y such that $1 = ax + by$. Deduce that if $a|bc$ with $\gcd(a, b) = 1$, then $a|c$.
28. Prove the relationship between gcd and lcm of two positive integers.
29. Solve the system of congruences $x \equiv 1 \pmod{3}, x \equiv 2 \pmod{5}, x \equiv 3 \pmod{7}$.
30. State Fermat's Theorem. Is its converse true? Justify your answer.
31. Prove that τ and σ are multiplicative functions.
32. Define a vector space. Verify whether the set $\mathbb{R}^2$ a vector space under the operations $(x_1, x_2) + (y_1, y_2) = (x_1 + y_1, 0)$ and $c(x_1, x_2) = (cx_1, 0)$.
33. Show that every linearly independent subset of a finite dimensional vector space can be extended to form a basis of the vector space.

34. Show that the linear mapping $f : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ given by $f(x, y, z) = (x + y + z, 2x - y - z, x + 2y - z)$ is both surjective and injective.
35. If the set $\{(1, 1, 0), (1, 0, 1), (0, 1, 1)\}$ is a basis of $\mathbb{R}^3$ and if $f : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ is linear such that $f(1, 1, 0) = (1, 2), f(1, 0, 1) = (0, 0), f(0, 1, 1) = (2, 1)$, determine f completely.

(5 × 9 = 45 marks)

Section D

*Answer any **one** question.
The question carries 15 marks.*

36. State and prove Division algorithm. Illustrate with an example.
37. Prove that the quadratic congruence $x^2 + 1 \equiv 0 \pmod{p}$, where p is an odd prime, has a solution if and only if $p \equiv 1 \pmod{4}$.
38. Let V and W be vector spaces over a field F and $f : V \rightarrow W$ be a linear map. Prove that :
- (a) Range (f) is a subspace of W .
 - (b) If V is spanned by a finite subset S of V , then Range (f) is spanned by $f(S)$.
 - (c) If X is a finite dimensional subspace of V such that $X \cap \ker f = \{0\}$ and if $B = \{v_1, v_2, v_3, \dots, v_n\}$ is a basis X , then $f(B)$ is a basis of $f(X)$.

(1 × 15 = 15 marks)